

Week 3: Monte Carlo simulation

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Abstract: The two-dimensional Ising model is exactly solved, so any sampler that claims to simulate it can be held to Onsager's answer. We model the spin system as a Markov chain and derive from detailed balance both the single-flip Metropolis rule and the Wolff cluster rule that flips whole domains at once. From the chain's samples we measure the magnetization curve, locate the critical temperature on finite lattices with an error bar that accounts for the chain's memory, and find that cluster flips remove the critical slowing down that defeats single flips at the transition. The evidence is a tested Rust tool, a run anyone can repeat, and a recording of the lattice as it heats. Together they show that a Monte Carlo estimate is only as good as its sampler, its finite-size extrapolation, and its error analysis.

The final checklist defines done. Every term in a Goal line is defined in the Part that delivers it.

Resources:

- Sandvik, *Monte Carlo simulations in classical statistical physics*, the PY502 lecture notes; each Part names its sections and pages.
- Susskind, *Statistical Mechanics*, lecture 9, the Ising model on video: spins, the energy, why the lattice orders, and what a phase transition is. Watch the first hour before the opening section; the mean-field part after it is background.

The Ising model

The Ising model is the course's reference problem, and every Part of this week samples it. It is the simplest system that has a phase transition: a magnet that holds its magnetization when cold and loses it at a sharp temperature when heated, as iron does at its Curie point. The same lattice, read differently, is a binary alloy ordering or a gas condensing on a lattice. It is the course's reference problem because it is *exactly solved*: there is a formula to check every number against, and a check that fails when the physics is wrong is what this course grades.

Open [the Ising lattice lab](#) in your browser. Heat it, cool it, and hover a spin as you read the model below.

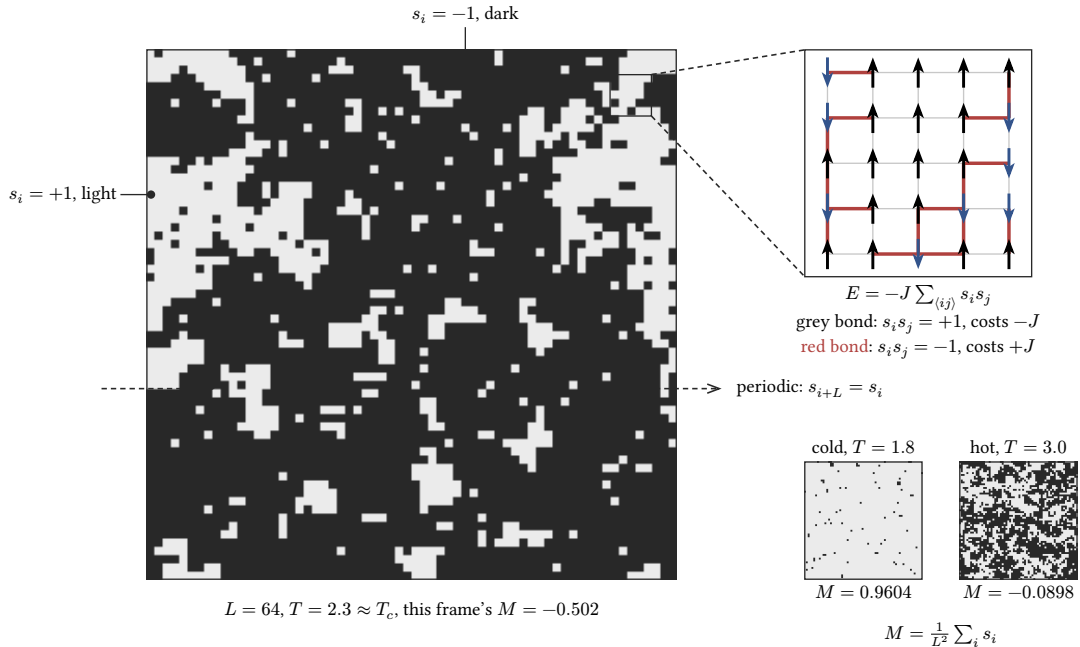


Figure 1: **The model on a real lattice.** A 64×64 configuration from the NumPy baseline at $T = 2.3$, light squares up and dark squares down, its edges glued to the opposite ones. The tile enlarges the framed window as arrows with their bonds: only red bonds, where neighbours disagree, cost energy. Below it, the same lattice cold and hot, with the magnetization of each frame.

Spins on a torus. Put a spin $s_i \in \{-1, +1\}$ on each site of an $L \times L$ square lattice: a tiny magnetic moment that points up or down, a light or dark square in Figure 1 and an arrow in its zoom. Periodic boundaries join each edge to the opposite edge, closing the lattice into a *torus* (a doughnut surface), so every site has exactly four neighbours and no site is at an edge. A *configuration* s is one assignment of all L^2 spins; for $L = 64$ there are 2^{4096} of them.

Energy. Neighbouring spins prefer to agree. The *energy* counts their disagreements:

$$E(s) = - \sum_{\langle ij \rangle} s_i s_j, \quad (1)$$

where the sum runs over nearest-neighbour pairs. An agreeing pair contributes -1 and a disagreeing pair $+1$. With N_{disagree} disagreeing bonds, $E = -2L^2 + 2N_{\text{disagree}}$. The energy per site is -2 when every spin agrees and approaches 0 for independent random spins. The coupling in front of the sum is set to $J = 1$, and every temperature in this sheet is measured in units of it.

Temperature. At temperature T , statistical mechanics gives configuration s the *Boltzmann probability*

$$P(s) = \frac{e^{-E(s)/T}}{Z}, \quad Z = \sum_s e^{-E(s)/T}. \quad (2)$$

Cold, only nearly aligned lattices carry significant weight. Hot, every configuration is about as likely as any other, so a hot lattice looks like noise. Z is a sum over all 2^{L^2} configurations that no computer evaluates; the methods of this week never need it.

Magnetization. The *magnetization* $m = \frac{1}{L^2} \sum_i s_i$ is the average spin, and measures how far the lattice has agreed on one side. Nothing in Equation 1 prefers up to down, so on a finite lattice the whole lattice flips over now and then, so the average of m tends to zero at every temperature. The average of $|m|$ remains positive at finite size, and it is what the ordered phase is measured by. Write that average as $\langle |m| \rangle$.

Onsager's exact solution. Onsager solved this model in 1944 [1]. Below a *critical temperature* the lattice picks a side, the *ordered* phase; above it thermal noise wins, the *disordered* phase. On an infinite lattice,

$$T_c = \frac{2}{\ln(1 + \sqrt{2})} = 2.26919, \quad \langle |m| \rangle = (1 - \sinh^{-4}(2/T))^{1/8} \text{ for } T < T_c, \quad (3)$$

and $\langle |m| \rangle = 0$ above T_c . On the infinite lattice the sign never flips, so the absolute value changes nothing there; it is a finite-size device. **Every measurement in this sheet is checked against this line.** A finite lattice rounds the kink at T_c , and its $\langle |m| \rangle$ stays a little above zero past T_c ; Part 2 measures that drift and removes it.

Domains and the correlation length. A *domain* is a connected region of spins that agree. Cold, one domain fills the lattice and only islands dissent. Hot, domains are small and short-lived. At T_c domains of every size coexist, from single sites to the whole lattice, because the *correlation length*, the distance over which spins still know about each other, has grown to the size of the lattice. Part 1 records this.

KEY CONCEPTS

- A *configuration* s assigns a *spin* $s_i \in \{-1, +1\}$ to every site of an $L \times L$ torus.
- The *energy* Equation 1 counts disagreeing neighbour pairs, with $J = 1$ as the unit of temperature.
- The *Boltzmann probability* Equation 2 weights each configuration by $e^{-E(s)/T}$; its normalization Z is never computed.
- The *magnetization* m is the average spin, and $\langle |m| \rangle$ is the observable that detects order on a finite lattice.
- Onsager's exact T_c and $\langle |m| \rangle$ curve Equation 3 separate the *ordered* phase from the *disordered* one and check every measurement of the week.
- A *domain* is a connected region of agreeing spins; at T_c the *correlation length* reaches the lattice size and domains of every size coexist.

Part 1: Monte Carlo ABC

Goal: a tested Rust sampler whose cold magnetization agrees with Onsager and whose hot mean $|m|$ is below 0.06, with runs reproducible from one seed and a recording the course viewer plays of domain changes through the transition.

① DO

Part 1 builds `ising`, a command that samples the model along one temperature ramp and records what it draws. Reuse the `AMAT5315-2026Fall-Exercise` repository from weeks 1 and 2, start the agent at its root, send blue panels to the agent, and run grey panels yourself in a second terminal. Runs you make to look at things go under `runs/`;

artifacts/ is reserved for the runs the instructor's gate reads, from Part 2 on. In the second terminal, create the week's folder and enter it:

```
> mkdir -p week3 && cd week3
```

Create a CLI contract file named `ising.design.toml` and copy the green panel below into it. It is the whole contract: one command, its arguments, and what it writes.

```
ising.design.toml
usage      = "ising --update metropolis --l 64 --t-from 1.5 --t-to 3.5 --t-step 0.05
--discard 2000 --measure 200 --every 20 --seed 2026 --out runs/ramp"
description = "sample the Ising model along a temperature ramp; no unstated defaults"

[args]
update = ""metropolis or wolff; one step is an l*l-proposal sweep or a cluster flip;
        metropolis picks sites independently and uniformly, with replacement""
l      = "integer lattice side, at least 2"

t-from = "lowest temperature; the ramp ascends from here, from an all-up lattice"
t-to   = "temperature upper bound; included only if reached by the temperature step"
t-step = "temperature step; each temperature starts from the previous lattice"

discard = "equilibration steps discarded at each temperature"
measure = "measured steps at each temperature"
seed     = "random seed; one random stream per run, carried through the ramp"
every    = ""record at measured steps every, 2*every, ... at each temperature;
0, the default, records none""

out = "output folder; may exist; overwrite same-named files written by this run"

[writes]
stdout = ""a header, then one tab-separated line per temperature:
        T, mean |M| over the measured steps, then the acceptance rate (metropolis),
        accepted flips per proposal over discard + measure,
        or the mean cluster size (wolff)""

"<out>/run.json" = ""L, update, t_grid, discard, measure, seed, sample_every = 1,
        time_unit = sweep for metropolis or cluster_flip for wolff""

"<out>/series.jsonl" = ""one object per measured step: L, T, sweep, M, E; 6 decimals;
        sweep starts at 1 after discard at each temperature;
        M is the signed mean spin; E is the energy per site;
        wolff rows add cluster_size, the spins flipped in that move""

"<out>/spins.jsonl" = ""frames when every > 0; each object: L, T, sweep, m, spins;
        sweep counts steps cumulatively across the whole ramp, including discard;
        spins is a row-major array of l*l integers, +1 for up and -1 for down""
```

The file is the tool: `usage` spells one complete run, `description` says what the command does, `[args]` gives the meaning of every flag, and `[writes]` fixes the printed table and the files the course gate reads. There is no nested command table, because `ising` has no subcommand. The model is not in the file: it goes in the prompt, which hands the contract to the agent:

```
Build `ising` in `week3/`, a Rust CLI sampling the two-dimensional Ising model: L by L
spins +1/-1 on a periodic square lattice, J = 1, no field, T in units of J. Implement
the command with its `metropolis` update as `week3/ising.design.toml` specifies.
Discuss uncertainties with me. Commit.
```

The agent will follow the contract to build the tool.

The model is fixed in the opening section; this Part builds the sampler that draws from it, and records what it draws.

The Boltzmann average. The quantity to measure is a weighted sum over every configuration,

$$\langle |m| \rangle = \sum_s |m(s)| P(s), \quad (4)$$

and for $L = 64$ the sum has 2^{4096} terms. A continuous variable would turn it into an integral over an exponentially large or even infinite space, where a grid of sample points fails for the same reason: the number of grid points grows exponentially with the dimension.

Importance sampling. Draw n configurations $s^{(1)}, \dots, s^{(n)}$ from a *sampling distribution* $Q(s)$ and average $|m(s)|P(s)/Q(s)$ over them; the ratio P/Q is the *importance weight* of a draw. For independent draws with finite variance, its statistical error falls as $1/\sqrt{n}$. The prefactor depends on the observable and on Q .

The Monte Carlo estimate. Uniform draws waste almost every sample: at low T the weight P sits on a vanishing fraction of the 2^{L^2} configurations, and a uniform draw almost never lands there (Figure 2). Sampling the target itself, $Q = P$, removes the importance weights. Every draw then counts equally, and

$$\langle |m| \rangle \approx \frac{1}{n} \sum_{k=1}^n |m(s^{(k)})|. \quad (5)$$

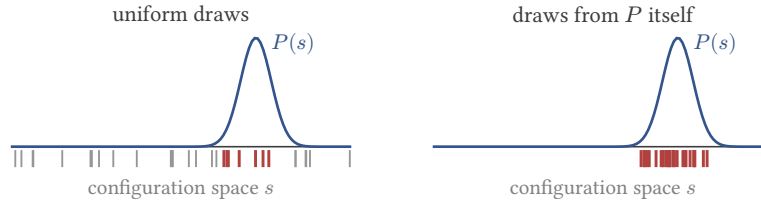


Figure 2: The same peaked target $P(s)$ over a one-dimensional cartoon of the configuration space, with thirty draws as ticks under the axis. Uniform draws (left) almost all fall where $P \approx 0$ and contribute nothing; draws from P itself (right) all land under the peak, in red.

So the key is to design a good sampler: a procedure that draws configurations with probability $P(s)$ itself. A *Markov chain* is such a sampler: a random walk whose next state depends only on the current one. Write $W(s \rightarrow s')$ for the probability of stepping from s to s' . This week builds two chains for the same P : single spin flips now, and cluster flips in Part 4.

Detailed balance and ergodicity. A distribution is *stationary* when one step of the walk leaves it unchanged: a crowd of walkers spread as P is still spread as P after everyone moves. *Detailed balance* is the sufficient condition that makes P stationary,

$$P(s)W(s \rightarrow s') = P(s')W(s' \rightarrow s), \quad (6)$$

the traffic from s to s' equals the traffic back, so no configuration gains or loses walkers. *Ergodicity* is the second condition: every configuration must be reachable from every other in a finite number of steps, so the chain cannot get stuck in a corner of the space (Figure 3) and the stationary distribution is unique. Convergence from any start also requires *aperiodicity*: the walk must not be locked into a repeating cycle of fixed length. Rejected Metropolis proposals provide self-transitions (a step that keeps the current configuration) at the finite positive temperatures used here. These conditions make the chain converge to P , and its states become the samples that Equation 5 needs. Two consequences run through the week: a run must first *equilibrate*, discarding the steps before the chain has converged, and how fast it converges is a property of the sampler, measured in Part 3 and improved in Part 4.

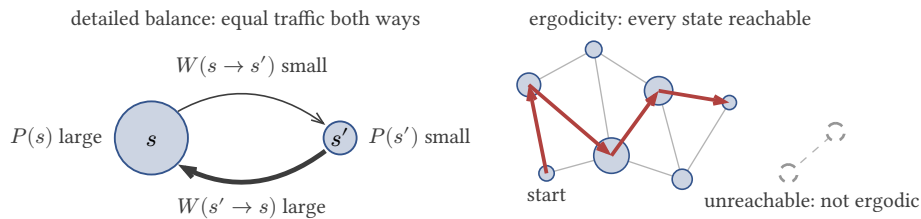


Figure 3: The chain in state space. Left: two configurations drawn with area proportional to P and arrows with weight proportional to W ; the thin arrow out of the likely state and the thick arrow into it carry equal traffic. Right: a connected graph of configurations with one walk of the chain in red; the dashed island no edge reaches is what ergodicity rules out.

The Metropolis rule. Pick a site uniformly at random and propose flipping its spin. Let ΔE be the energy change the flip would cause. Accept the flip with probability

$$A = \min(1, e^{-\Delta E/T}), \quad (7)$$

and otherwise keep the old configuration. One *sweep* is L^2 such proposals, accepted or not, and simulation time is counted in sweeps. Downhill proposals always pass; uphill proposals pass with a temperature-dependent coin flip. Substituting Equation 7 into Equation 6 leaves only the ratio $P(s')/P(s) = e^{-\Delta E/T}$, in which Z cancels: the sampler never needs Z , only energy *differences*.

The energy change of one flip. Flipping s_i changes only the four terms of Equation 1 that contain site i .

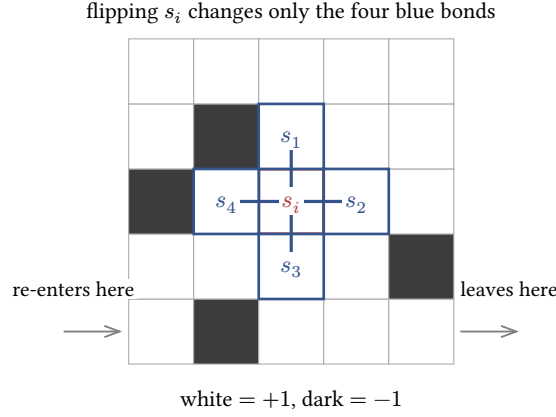


Figure 4: A 5×5 patch of the lattice with the proposed site s_i in red and its four neighbours $s_1, \dots, s_4$ in blue. Only the four blue bonds change energy when s_i flips. Periodic boundaries join the right edge to the left edge and the top to the bottom.

Flipping s_i replaces each of those four bond terms $-s_i s_j$ by $+s_i s_j$, so

$$\Delta E = 2s_i(s_1 + s_2 + s_3 + s_4). \quad (8)$$

Each neighbour sum $s_1 + s_2 + s_3 + s_4$ is one of $-4, -2, 0, 2, 4$, so ΔE takes one of five values. Three give acceptance 1, so Equation 7 has only three distinct acceptance probabilities at a fixed finite T . Computing them once into a five-entry table replaces one exponential per proposal with one array lookup.

ΔE	-8	-4	0	+4	+8
A at $T = 2.3$	1	1	1	0.1757	0.0309

Energy histograms. An energy histogram counts how many recorded configurations lie in each energy interval. For two temperatures $T_a < T_b$, the *log ratio* is $\ln(P_{T_b}(E)/P_{T_a}(E))$, the hot histogram divided by the cold one. For two temperatures on the same lattice, the number of states at each energy cancels in their probability ratio. The normalization contributes only a constant:

$$\ln(P_{3.1}(E)/P_{3.0}(E)) = (1/3.0 - 1/3.1)E + \text{const}. \quad (9)$$

The expected slope is 0.0107527 per energy unit. Use total energy on the horizontal axis. Retain bins containing at least five observations in both histograms. Their points fluctuate because the histograms come from finite, correlated samples. This comparison checks the relative temperature dependence over the shared energy range.

The ramp recording. The usage line of the contract runs one ramp through the whole transition and records frames of the lattice, because **what changes at the transition is the size of the domains, not the number of up spins; the lattice shows it directly**. Each temperature starts from the final lattice of the previous one, a *warm start*, which saves reaching equilibrium from all-up every time. It does not replace equilibration: the run discards 2000 sweeps at every temperature, and a ramp that discards only 200 lags behind as it crosses the critical region and shows an ordered lattice at $T = 2.3$ with $m \approx 0.78$. The ten frames kept per temperature illustrate the transition; they do not establish equilibrium at every temperature, and Part 3 measures how long the chain retains correlations. The frames file is [JSON Lines](#), one frame per line with the keys the contract fixes, and its sweep counter runs across the whole ramp so the viewer's history reads as one timeline.

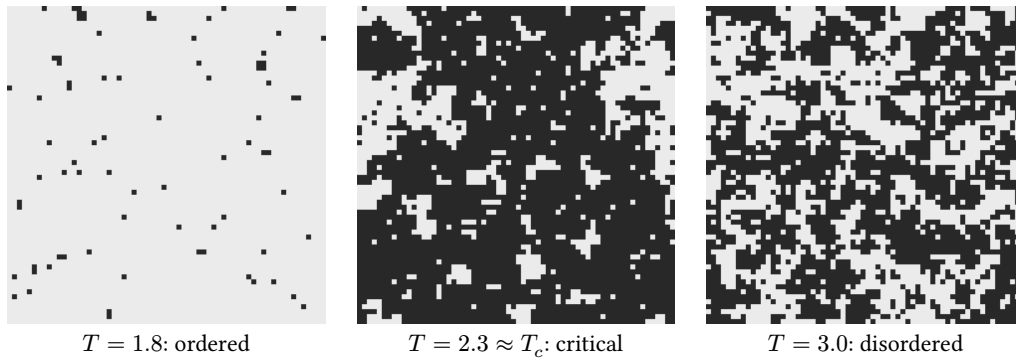


Figure 5: Equilibrated 64×64 lattices at three temperatures, white for $+1$ and dark for -1 , from the reference run behind every figure in this sheet. Cold: one domain fills the lattice and only islands dissent. Near T_c : domains of every size coexist, from single sites to the whole lattice. Hot: no domain outlives a few sweeps and $m \approx 0$.

- Read [Sandvik’s notes](#), section 2.2 “Lattice and spin models”, pages 4–6. Identify the Ising energy and what the sum over nearest neighbours means.
- Read section 3, “Importance sampling and detailed balance”, pages 7–10, and section 4, “Metropolis algorithm for the Ising model”, pages 10–15. Find what a Markov chain is, what detailed balance and ergodicity require of it, and where detailed balance is used to derive the acceptance probability [Equation 7](#).
- Skim the [Rust Rand Book](#), “Seeding RNGs”. Explain what a seed does and why a fresh random seed would make this week’s checks impossible.
- Read Sandvik’s notes, section 5, page 18, Figure 8: three temperatures below, at, and above T_c on an $L = 32$ lattice. Compare its domain sizes with the ones above.

KEY CONCEPTS

- The *Monte Carlo estimate* [Equation 5](#) averages magnetization over sampled lattices.
- The *Metropolis rule* [Equation 7](#) samples Boltzmann weights using the energy change [Equation 8](#).
- The *histogram log ratio* [Equation 9](#) tests relative Boltzmann weights through its slope.
- A *seed* fixes the random stream so that the same settings reproduce the same measurements.
- *Domains* are connected regions of agreeing spins whose sizes change across the transition.
- A *frame* records one lattice for comparison with the domain patterns in [Figure 5](#).

3 VERIFY

Terminal: inside week3/ , then your browser. You run these yourself: a check outsourced to the thing being checked proves nothing. The tool is run by name: after the agent’s last commit of a Part, `cargo install --path .` builds it in release mode, the optimized build, and copies `ising` into the folder your shell searches for commands, which `rustup` set up, so every check below starts from that line. Deliverables: the printed lines at two temperatures, the Boltzmann chart, and the three stamped viewer frames.

```
> cargo install --path . --quiet
```

- (1) **The magnetization is right at two temperatures.** A ramp of one temperature is a single run from all-up; one frame at the end is enough here.

```
> ising --update metropolis --l 64 --t-from 1.8 --t-to 1.8 --t-step 0.1 \
  --discard 2000 --measure 2000 --seed 2026 --out runs/T1.8
  ising --update metropolis --l 64 --t-from 3.0 --t-to 3.0 --t-step 0.1 \
  --discard 2000 --measure 2000 --seed 2026 --out runs/T3.0
```

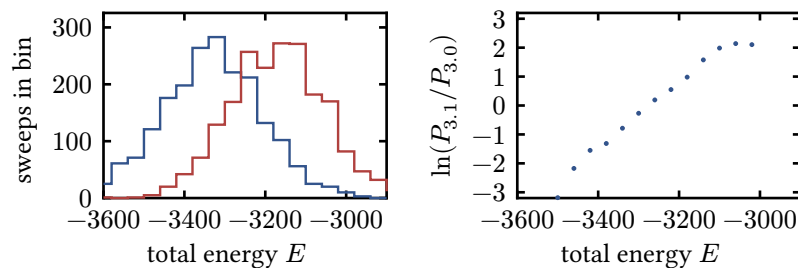
Each run prints a header and one line: temperature, mean $|M|$, acceptance rate. Expected: the first reports a mean $|M|$ between 0.95 and 0.965, with about 4% of proposals accepted; the answer key prints `0.9568` and `0.0410`, agreeing with [Equation 3](#) to four decimals. The second reports a mean $|M|$ below 0.06 with about 46% accepted; the answer key prints `0.0414` and `0.4601`. A result outside these windows needs investigation. Check the energy convention, temperature, acceptance rule, and equilibration before assigning a cause.

- (2) **The chain samples the Boltzmann distribution.** One more run, one temperature step above the hot one:

```
> ising --update metropolis --l 64 --t-from 3.1 --t-to 3.1 --t-step 0.1 \
  --discard 2000 --measure 2000 --seed 2026 --out runs/T3.1
```

Draw histograms of the total energy, ``E`` times 4096 per row of ``series.jsonl``, for the runs in ``week3/runs/T3.0/`` and ``week3/runs/T3.1/``, with bins 40 units wide. Plot their log ratio, second over first, where both bins hold at least five rows. Overlay slope $1/3.0 - 1/3.1$. Save ``week3/evidence/boltzmann.png`` and keep its drawing command. Commit.

Open `evidence/boltzmann.png` and compare with this reference. Pass: the two histograms overlap, and the log-ratio points increase along the dashed line of slope 0.0107527 per energy unit from Equation 9, each within about 0.5 of it where a bin holds more than 20 sweeps (sparser bins scatter more). A systematic departure calls for more samples and checks of the energy and temperature conventions. Agreement supports the Boltzmann ratio over the overlap; it does not test every configuration probability.



Energy histograms and their log ratio at $L = 64$ from the answer key's runs, 2000 measured sweeps each. Left: $T = 3.0$ in blue, $T = 3.1$ in red, the hot run about 150 units higher in energy. Right: the ratio of the two histograms on a log scale is a straight line of the slope Boltzmann predicts, dashed.

- (3) **One seed, one run.**

```
> ising --update metropolis --l 64 --t-from 1.8 --t-to 1.8 --t-step 0.1 \
  --discard 2000 --measure 2000 --seed 2026 --out runs/a
  ising --update metropolis --l 64 --t-from 1.8 --t-to 1.8 --t-step 0.1 \
  --discard 2000 --measure 2000 --seed 2026 --out runs/b
  diff runs/a/series.jsonl runs/b/series.jsonl && echo IDENTICAL
  ising --update metropolis --l 64 --t-from 1.8 --t-to 1.8 --t-step 0.1 \
  --discard 2000 --measure 2000 --seed 2027 --out runs/c
```

Expected: both seed-2026 runs succeed and save identical nonempty series; `diff` prints differing lines, or nothing when the files match; `&& runs echo IDENTICAL` only when they match. The seed-2027 mean differs in its last digits while staying inside the same window; the answer key gives 0.9574 there. If seed 2027 reproduces seed 2026 exactly, the generator is ignoring its seed. The same seed gives the same line on any machine, because the generator is fixed by the code and not by the hardware, so a classmate can reproduce your number.

- (4) **Heating breaks up the ordered domain.** The usage line of the contract runs the ramp with frames, a few seconds; commit the recording so the viewer can load its public address.

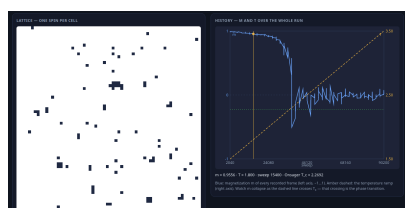
```
> ising --update metropolis --l 64 --t-from 1.5 --t-to 3.5 --t-step 0.05 \
  --discard 2000 --measure 200 --every 20 --seed 2026 --out runs/ramp &&
  test -s runs/ramp/spins.jsonl &&
  cp runs/ramp/spins.jsonl spins.jsonl &&
  git add spins.jsonl && git commit -m "week 3 ramp" && git push
```

The recording follows the `spins.jsonl` format the contract fixes. On GitHub, open the committed `week3/spins.jsonl`, select **Raw**, and copy its address from the browser. Open the [course viewer](#), paste the address into *Recording URL*, and press *Load*. The page draws the lattice as one square per spin beside the history

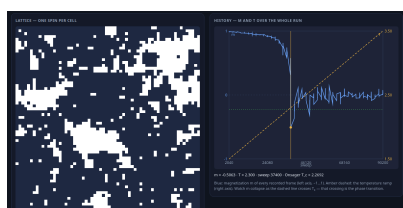
of m against sweep, with the temperature ramp on a second axis; a local recording can be dropped on the page instead.

Choose 1.8, 2.3, and 3.0 in *Temperature* in turn. For each, use *Copy link* and open that complete link in a private browser window. Pass: the page loads without a login and shows the requested temperature. Compare the domain patterns with the three references: predominantly one colour when cold, large connected regions near the transition, and smaller regions when hot. Either colour may dominate because reversing every spin leaves the energy unchanged.

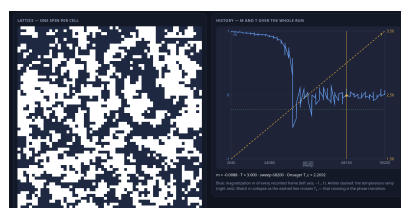
Read all ten frames at each temperature. They illustrate fluctuations; a single ordered-looking frame near T_c does not show failed equilibration. Judge the numerical magnetization from the longer averages in this Part and Part 2, using $|m|$. Press *save PNG* for one representative frame at each temperature; it writes the frame with its temperature and magnetization printed on it, the *stamp*. Save them in `week3/evidence/` as `viewer-T1.8.png`, `viewer-T2.3.png`, and `viewer-T3.0.png`.



$T = 1.8$: one domain, island dissenters



$T = 2.3$: domains of every size



$T = 3.0$: no surviving domain

Extension: the ramp that runs backwards. Record a descending ramp from $T = 3.5$ to $T = 1.5$ that starts from random spins instead of all-up. Below T_c it can freeze into a striped state, two domains wrapped around the torus, which single-spin flips cannot dissolve in any reasonable time. Find the lowest temperature at which your descending ramp still agrees with the ascending one, and say what that says about the anneal direction this sheet fixed.

Part 2: The magnet dies

Goal: a magnetization curve compared with Onsager's infinite-lattice result that shows finite-size rounding, and a critical temperature measured from two sizes within 2% of his exact value, with the raw rows that back both.

① DO

Part 1 built the command; this Part runs it four times, on two sizes and two temperature grids, with every value spelled out. Nothing is added to the contract. Install the tool and start the four ramps in the second terminal; together they take about two minutes:

```
> cargo install --path . --quiet
ising --update metropolis --l 32 --t-from 1.5 --t-to 3.5 --t-step 0.1 \
  --discard 2000 --measure 5000 --seed 1042 --out artifacts/coarse-l32
ising --update metropolis --l 64 --t-from 1.5 --t-to 3.5 --t-step 0.1 \
  --discard 2000 --measure 5000 --seed 42 --out artifacts/coarse-l64
ising --update metropolis --l 32 --t-from 2.0 --t-to 2.6 --t-step 0.05 \
  --discard 2000 --measure 100000 --seed 1042 --out artifacts/window-l32
ising --update metropolis --l 64 --t-from 2.0 --t-to 2.6 --t-step 0.05 \
  --discard 2000 --measure 100000 --seed 42 --out artifacts/window-l64
```

While they run, ask for the two charts and the critical-temperature estimate, both scripts over the rows the ramps write; `artifacts/` and `runs/` stay out of git:

Read the metropolis runs in ``week3/artifacts/`` and draw two plots using matplotlib. Plot mean $|M|$ against temperature for $L = 64$ alongside Onsager's infinite-lattice curve. Plot susceptibility ($\chi(T) = L^2 (\text{mean}(M^2) - \text{mean}(|M|)^2) / T$) against temperature for both sizes. Save ``week3/evidence/magnetization.png`` and ``week3/evidence/susceptibility.png``. Keep the script that draws them, and run it once the ramps have finished. Discuss uncertainties with me. Commit.



Write `week3/scripts/peaks.py`, reading the metropolis runs in `week3/artifacts/`. Use the same definition of susceptibility `chi`. Fit a quadratic to the five points around each size's largest `chi`, and take its maximum as `T_peak`. Report `T_c = 2 * T_peak(64) - T_peak(32)`, and mean `|M|` at the lowest temperature of each size. Print the values and save them as `week3/evidence/peaks.txt`. Run it once the ramps have finished. Discuss uncertainties with me. Commit.

② UNDERSTAND

This Part measures $\langle |m| \rangle$ at every temperature and compares the curve with Onsager's exact result, Equation 3.

The sweep protocol. One run covers two lattice sizes so that the finite-size effect can be measured rather than assumed. Its temperature grid is finer inside the *critical window*, the range $2.0 \leq T \leq 2.6$ where the transition happens, and its sampling is longer there, because the chain loses its memory slowly at the transition and Part 3 measures how slowly.

Given	Value
Sizes	$L = 32$ and $L = 64$.
Temperature grid	1.5 to 3.5 in steps of 0.1, refined to steps of 0.05 between 2.0 and 2.6: 27 temperatures.
Equilibration	2000 sweeps at each temperature, discarded.
Measurement	5000 sweeps at each temperature; 100000 at the 13 temperatures between 2.0 and 2.6. Record after every sweep.
Seeds	42 for $L = 64$, 1042 for $L = 32$.
Cost	About 7×10^9 proposals: two minutes in a release build, and 160 MB of recorded rows.

The artifact contract. Two files per run, written into that run's folder. The instructor's gate reads every folder under `artifacts/`, merges the Metropolis runs, and recomputes the physics itself, so the field names are fixed.

File	Required fields
<code>run.json</code>	<code>L</code> , <code>update</code> as "metropolis", <code>t_grid</code> as the list of temperatures, <code>discard</code> , <code>measure</code> , <code>seed</code> , <code>sample_every</code> equal to 1, and <code>time_unit</code> as "sweep".
<code>series.jsonl</code>	One JSON object per measurement sweep: <code>L</code> , <code>T</code> , <code>sweep</code> as the measurement-sweep index at that temperature, <code>M</code> as the signed mean spin, which is m , and <code>E</code> as the energy per site. Round <code>M</code> and <code>E</code> to 6 decimals.

Within a folder the rows appear in ramp order, every temperature ascending. 160 MB of them across the four folders must stay out of git.

Susceptibility and the finite-size shift. The *susceptibility* measures the size of the magnetization's fluctuations,

$$\chi(T) = \frac{L^2(\langle m^2 \rangle - \langle |m| \rangle^2)}{T}, \quad (10)$$

This *absolute-magnetization susceptibility* uses $\langle |m| \rangle^2$ to remove reversals between the two ordered orientations. The zero-field response susceptibility instead subtracts $\langle m \rangle^2$. Use the former consistently for this sheet's peak-location comparison. On an infinite lattice that peak would sit exactly at T_c and be infinitely tall. **On a finite lattice the correlation length cannot exceed L , so the peak is rounded. Its leading shift scales as $1/L$.** Two sizes can cancel that leading term. They cannot test the scaling assumption or remove higher-order corrections. Fit a parabola through the five grid points centred on the largest χ and take its vertex as $T_{\text{peak}}(L)$; then

$$T_c \approx 2T_{\text{peak}}(64) - T_{\text{peak}}(32) \quad (11)$$

cancels the leading $1/L$ term, because $T_{\text{peak}}(L) \approx T_c + a/L$ and $L = 64$ has half the error of $L = 32$.

The gate's conditions. The course gate recomputes these from your raw rows:

Condition	Measured quantity	Pass condition
Ordered phase	$\langle m \rangle$ at the lowest temperature, each size	≥ 0.9
Critical temperature	$\frac{ T_c - 2.26919 }{2.26919}$ with T_c from Equation 11	< 0.02

- Read Sandvik’s notes, section 4.2, “Evaluating physical observables”, pages 13–15. Find how the susceptibility is computed from the magnetization of the sampled configurations.
- Read section 7, “Critical phenomena and Finite-size scaling”, pages 27–33. Find why a finite lattice rounds the transition and how the peak position moves with L .

The transition is located from the raw rows by a script, not by the sampler; VERIFY asks for it.

KEY CONCEPTS

- The *sweep* measures $\langle |m| \rangle$ at 27 temperatures and two sizes, finer and longer inside the *critical window* $2.0 \leq T \leq 2.6$.
- The *artifact contract* is two files, the settings and one row per measurement sweep, read by a grader who never sees your code; the raw rows exist because Part 3 re-examines them.
- The *susceptibility* Equation 10 peaks where the lattice is most undecided; on a finite lattice the peak sits above T_c by about $1/L$.
- Two sizes cancel that drift through Equation 11, which turns the two peaks into the claim: one critical temperature within 2% of Onsager’s.
- The *gate* recomputes the ordered phase and that critical temperature from your rows.

3 VERIFY

Terminal: inside week3/. Deliverables: the magnetization chart, the susceptibility chart, and the extrapolated critical temperature.

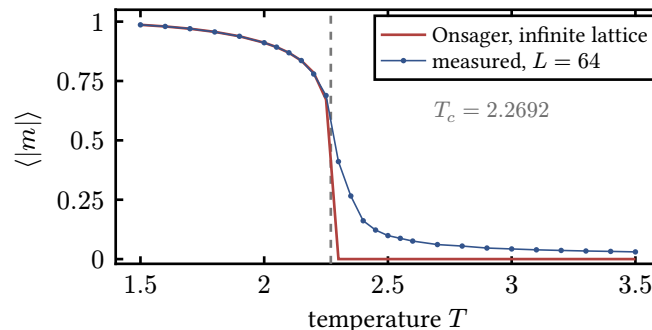
- (1) **A finite lattice rounds the magnetization transition.** The ramps DO started have finished:

```
> ls artifacts/*/run.json artifacts/*/series.jsonl
```

Each of the four folders holds its `run.json`, recording the size, seed and grid, and its `series.jsonl` in the artifact format above.

Open `evidence/magnetization.png` and compare with this reference.

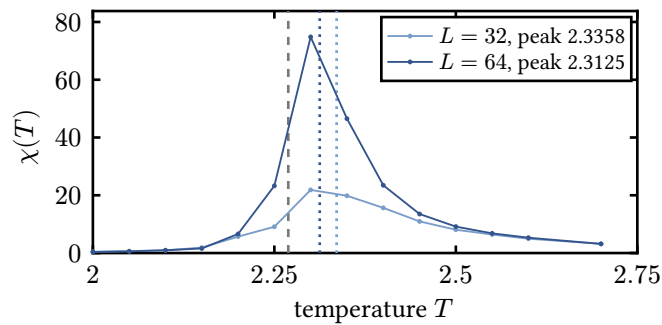
Pass: at $T = 1.5$ each size has mean $|m| \geq 0.9$. At $L = 64$, $T = 3.0$, the mean is below 0.06. Below the transition the curve approaches the infinite-lattice result. Near T_c , finite-size rounding permits visible departures on either side of the transition. The reference $L = 64$ curve reads about 0.69 at $T = 2.25$, 0.41 at 2.3 and 0.08 at 2.6, and yours should sit within about 0.1 of those.



Magnetization against temperature: the contract run at $L = 64$, NumPy reference implementation, against Equation 3. The dashed line is T_c . The finite lattice rounds the transition, so the measured points sit above the exact curve near T_c and leave a tail above it.

- (2) **The susceptibility peaks drift with size.** Open `evidence/susceptibility.png` and compare with this reference. Pass: both sizes peak between 2.30 and 2.40, the $L = 64$ peak is taller and narrower than the $L = 32$

peak, and it sits at a lower temperature. A shared grid maximum can occur when the grid cannot resolve the shift. Compare the fitted peak positions and their uncertainty before interpreting it.



Susceptibility against temperature: both sizes from the contract run, NumPy reference implementation, with the dashed line at T_c and the dotted lines at the two fitted peaks. The peak moves down and grows as L doubles.

(3) The critical temperature lands within 2% of Onsager's.

```
> cat evidence/peaks.txt
```

Expected: two ordered-phase values above 0.9, the limit $m \rightarrow 1$ as $T \rightarrow 0$ that the gate also checks (the answer key prints 0.9866 and 0.9865); two fitted peaks within 0.03 of 2.33 for $L = 32$ and 2.31 for $L = 64$, and a critical temperature within 2% of 2.26919. The answer key prints

$T_c = 2.2779$ (peaks: L32 2.3346, L64 2.3062), a deviation of +0.38%; the chart above, from the NumPy baseline, fits its peaks at 2.3358 and 2.3125, the same pair to within that width. These are example values from one random stream. Part 3 estimates the uncertainty in your fitted peaks. A discrepancy can arise from sampling, equilibration, the peak fit, or the temperature convention; its size alone does not identify the cause.

Part 3: How much to trust Part 2

Goal: correlation plots that explain the uncertainty in magnetization, and a critical-temperature estimate with a sampling error whose sensitivity to block length is reported.

① DO

Nothing is added to the contract: this Part is a study of the rows Part 2 already wrote. Two of its computations take a while, the autocorrelation times over every temperature and a block bootstrap of the peaks; send both now and read on:

Write `week3/scripts/errors.py`, reading the metropolis runs in `week3/artifacts/`. Print one line per size and temperature: L , T , mean $|M|$, the naive standard error over the measured steps, the standard error from 50 block averages, their ratio, and the integrated autocorrelation time, summed until the lag exceeds six times the sum. Run it into `week3/evidence/errors.txt`. Discuss uncertainties with me. Commit.

Block-bootstrap the window runs in `week3/artifacts/` at block lengths 2000, 4000, and 8000, 500 replicates each. Draw the susceptibility for both sizes with the five-point parabola fits, shading the envelope of the 500 fitted parabolas. Mark `2.26919`. Save `week3/evidence/chi-bootstrap.png`. Discuss uncertainties with me. Commit.

② UNDERSTAND

The critical temperature of Part 2 came with no error bar. This Part estimates its sampling error from the saved rows. It also checks whether the result depends on how those rows are grouped into blocks.

The autocorrelation function. One sweep proposes L^2 single flips and accepts a few percent of them near $T = 1.8$, so the lattice after a sweep is mostly the lattice before it. Successive measurements are therefore not independent draws from Equation 2; they are a correlated time series. The *autocorrelation function* measures how long the chain remembers:

$$\rho(t) = \frac{\langle a_k a_{k+t} \rangle - \langle a_k \rangle^2}{\langle a_k^2 \rangle - \langle a_k \rangle^2}, \quad (12)$$

where a_k is the measurement after sweep k , here $|m|$, and t is the lag in sweeps. It starts at $\rho(0) = 1$ and decays.

The integrated autocorrelation time collects the whole decay into one number:

$$\tau_{\text{int}} = \frac{1}{2} + \sum_{t \geq 1} \rho(t). \quad (13)$$

The sum is truncated in practice, because at large t the estimated $\rho(t)$ is noise: the standard cut stops it once the lag exceeds six times the running total, which is the rule this Part's script uses. With τ_{int} in hand, n correlated measurements are worth

$$n_{\text{eff}} = \frac{n}{2\tau_{\text{int}}} \quad (14)$$

independent ones, and the error bar on their mean is

$$\sigma_{\text{honest}} = \sigma_{\text{naive}} \sqrt{2\tau_{\text{int}}}, \quad \sigma_{\text{naive}} = \frac{s}{\sqrt{n}}, \quad (15)$$

with s the standard deviation of the measurements. Quoting σ_{naive} where τ_{int} is large understates the uncertainty by the factor $\sqrt{2\tau_{\text{int}}}$.

Binning. Cut the series into blocks of equal length, average each block, and take the standard error of those block means. While the block is shorter than τ_{int} , neighbouring blocks are still correlated and the estimate is too small. Once the block is several times τ_{int} , the block means are effectively independent and the estimate stops growing. That flat region is the *plateau*, and its height is the honest error bar. A fixed count of fifty blocks does not establish a plateau. At 100000 sweeps, each block has length 2000. The reference estimate $\tau_{\text{int}} = 669.2$ makes that only $2.13\tau_{\text{int}}$. Compare several longer block lengths and report how many blocks remain. If the estimate still changes or too few blocks remain to locate a plateau, report that the run does not determine a stable error bar.

Critical slowing down. The decay length of Equation 12 is not a constant of the model; it explodes at the transition and grows with the lattice. Single flips must dissolve the domains of Figure 5 one spin at a time, so the number of sweeps needed grows as L^z , with the *dynamic exponent* $z \approx 2.2$ for this update rule. That is why Part 2 spends twenty times more sweeps inside the critical window than outside it. A chart of τ_{int} against T needs a *logarithmic axis*, on which each tick multiplies the value by ten, to show the flat parts and the spike together.

The block bootstrap. A *block bootstrap* creates a new series by drawing consecutive blocks with replacement, so a block may be drawn more than once. It preserves correlations inside each block. Resample each temperature and size separately within the critical window $2.0 \leq T \leq 2.6$. Recompute $\chi(T)$, both five-point peak fits, and Equation 11 for every bootstrap replicate. The standard deviation of the replicated critical temperatures estimates its sampling error. Compare lengths 2000, 4000, and 8000 sweeps using 500 replicates each. Report fits whose parabolas do not bend downward, or whose peaks fall outside the five fitted temperatures. These are failed fits; keep their count in the report. Call an error *stable* when its values at the three block lengths agree within a tenth of their mean; otherwise it is *unresolved* and is reported as such. Stability across block lengths supports the estimate. This error excludes finite-size corrections and the bias of fitting five grid points: a peak is not a parabola, so a parabola through five points of an asymmetric peak puts its vertex slightly off the true maximum, by an amount more sweeps do not shrink.

- Read Sandvik's notes, section 5, "Autocorrelation functions", pages 15–19. Find the definition of the integrated autocorrelation time and its behaviour at the critical temperature.
- Read the binning discussion in section 4.2, pages 13–15, and the warning about bin length on page 19. Find the condition a bin must satisfy for the standard deviation of bin averages to be an honest error bar.

KEY CONCEPTS

- The *autocorrelation function* Equation 12 measures persistence between successive measurements.
- The *integrated autocorrelation time* Equation 13 sets the error correction $\sqrt{2\tau_{\text{int}}}$ in Equation 15.
- *Binning* estimates the mean's error from block means, with a plateau supporting a stable estimate.
- *Critical slowing down* makes τ_{int} peak near the transition and grow with lattice size.
- The *block bootstrap* carries sampling uncertainty through the peak fits but excludes finite-size and fitting bias.

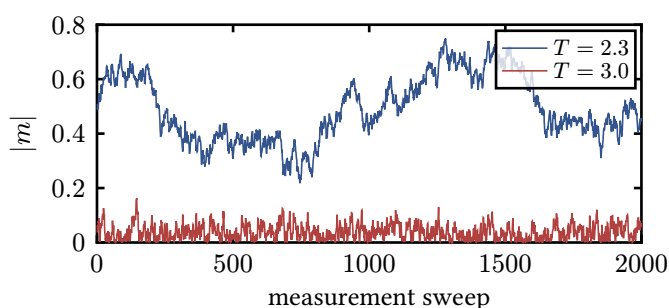
Terminal: inside week3/ . Deliverables: the autocorrelation-and-binning plot, the autocorrelation-time chart, and the critical temperature with its sampling error.

```
> cargo install --path . --quiet
```

(1) **Successive sweeps are alike.** Look at the raw series before any statistic:

Draw the absolute magnetization over the first 2000 recorded sweeps at temperatures 2.3 and 3.0, for size 64 of the metropolis runs in `week3/artifacts/`. Save it as `week3/evidence/trace.png` and keep the command that draws it. Discuss uncertainties with me. Commit.

Open `evidence/trace.png` . Pass: at $T = 3.0$ the trace jitters around 0.04 with much shorter correlations; at $T = 2.3$ it drifts in excursions that last hundreds of sweeps and can sit far from the long-run mean of about 0.42 for the whole window, as below. The trace illustrates persistence. Count effective samples with Equation 14 rather than by identifying excursions by eye.



Absolute magnetization sweep by sweep at $L = 64$ from the answer key's window run, the first 2000 measured sweeps. The hot trace is noise around its mean; the critical trace remembers where it was for hundreds of sweeps.

(2) **Correlations enlarge the error much more near the transition.** Read the three $L = 64$ rows from the table DO's first script wrote:

```
> awk 'NR == 1 || ($1 == 64 && ($2 == 1.5 || $2 == 2.3 || $2 == 3.5))' \
    evidence/errors.txt
```

Expected: a header and three data rows. At $T = 2.3$, the 50-block error is at least 10 times the naive error, and τ_{int} is hundreds of sweeps. At $T = 1.5$ and 3.5, the error ratio is about 2, and τ_{int} is a few sweeps.

At $T = 2.3$, the answer key gives an error ratio of 31.16, $\tau_{\text{int}} = 669.22$, and a mean of 0.4225. At $T = 1.5$, it gives an error ratio of 2.16 and $\tau_{\text{int}} = 1.61$. At $T = 3.5$, it gives an error ratio of 2.05 and $\tau_{\text{int}} = 2.33$. These are examples from one random stream, not exact targets.

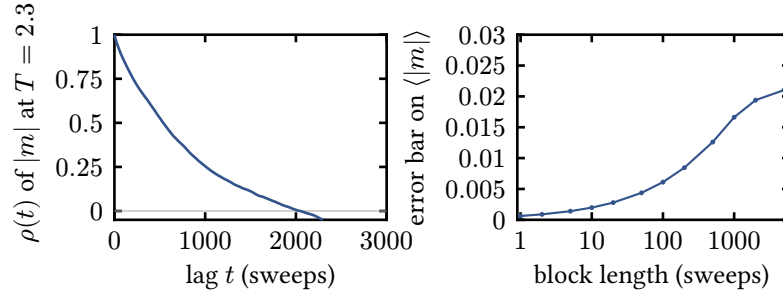
The 50-block estimate is provisional. Each block of the 100000-sweep series contains 2000 sweeps, about three autocorrelation times in the reference run. Equation 15 predicts an error ratio of $\sqrt{2} \times 669.2 = 36.6$, larger than the measured 31.16. The reference run's blocks are too short to reach the plateau, so its blocked error is still an underestimate. The next check examines this block-length dependence.

(3) **The honest error is still growing at fifty blocks.**

Draw two panels for size 64 at temperature 2.3 from the metropolis runs in `week3/artifacts/`: autocorrelation of $|M|$ against lag, and the standard error of mean $|M|$ against block length. Put block length on a logarithmic axis. Save `week3/evidence/acf-binning.png` and keep its command. Discuss uncertainties with me. Commit.

Open `evidence/acf-binning.png` . The block-length-one point must equal the naive error in `evidence/errors.txt` ; longer blocks should expose the effect of correlations. Compare the increase with this

reference, drawn from the answer key's run under the same update rule and protocol. A plateau supports an error estimate. The reference has none: its curve still rises at blocks of 5000 sweeps, where only 20 blocks remain, from 0.0006 at length one to 0.021, so the honest error at this temperature is at least 0.021 and unresolved. Expect the same of your run, and record "sampling error unresolved" unless your curve flattens.



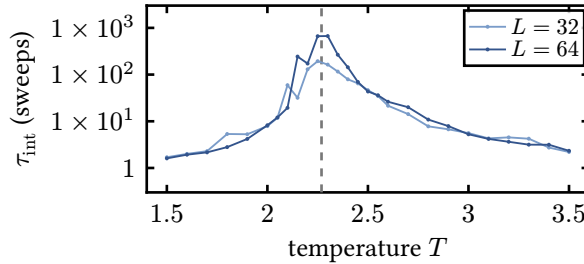
Autocorrelation and binning of $|m|$ at $L = 64$, $T = 2.3$, from the answer key's window run, $\tau_{\text{int}} = 669.2$ sweeps. Right: the binning estimate of the error bar on $\langle |m| \rangle$ against block length; the leftmost point is σ_{naive} .

(4) The autocorrelation time spikes where the physics is.



Draw the integrated autocorrelation time against temperature for both sizes from the metropolis runs in `week3/artifacts/`. Use a logarithmic vertical axis. Save it as `week3/evidence/tau.png` and keep the command that draws it. Discuss uncertainties with me. Commit.

Open `evidence/tau.png` and compare with this reference. Pass: both curves are flat and of order one sweep away from T_c , rise by more than a factor of a hundred near it, and the $L = 64$ spike is taller than the $L = 32$ spike; the answer key peaks at about 670 sweeps for $L = 64$ and 190 for $L = 32$, and yours should peak within a factor of two of those.



Integrated autocorrelation time against temperature, both sizes, logarithmic axis, computed with Equation 13 from the answer key's runs. The dashed line is T_c . Away from it, successive sweeps are nearly independent; at it the chain slows, and it slows more on the larger lattice.

(5) The critical-temperature claim carries a sampling error. Open `week3/evidence/chi-bootstrap.png`, which DO's second prompt drew. Pass: each envelope is narrow and sits above the exact temperature. Compare the envelope width across the three block lengths; widths that differ by more than a tenth of their mean make the sampling error unresolved, so report it as unresolved rather than quoting a precision. The central estimate still faces Part 2's 2% comparison with 2.26919; its sampling error does not include finite-size bias.

The Rust reference's window runs give $T_c = 2.2771$ with bootstrap standard errors 0.0096, 0.0095 and 0.0092 at the three block lengths, within a tenth of their mean, so its sampling error is resolved at about 0.01. The finite-size bias of +0.008 is not inside that bar; the two-size extrapolation, not more sweeps, is what would shrink it.

Extension: how long is long enough. From the τ_{int} your run reports at $L = 64$, $T = 2.3$, compute n_{eff} with Equation 14 for the 100000 recorded sweeps. Then estimate the longer run needed to make the correlation-corrected error as small as today's naive error, assuming τ_{int} stays fixed. Convert that run length to hours using your measured sweep rate. Report both numbers in `week3/README.md`.

Part 4: Beat critical slowing down

Goal: a cluster sampler with an equilibrium-magnetization comparison that states whether agreement is provisional, a critical temperature within 2% of the exact value, and reduced spin-update work per independent sample near the transition.

① DO

Single flips crawl near the transition, so this Part builds the second update rule. Nothing is added to the contract:

--update already names `wolff`, and its line already says that one step is then one cluster flip.



Add the `wolff` value of `--update` to `ising` in `week3/`, as `week3/ising.design.toml` specifies. Discuss uncertainties with me. Commit.

Read the agent's summary and resolve its open questions. Once it has committed, install and start the window with cluster flips on both sizes, then read on; the reference run took about 95 seconds on an Apple M2 in a release build:



```
cargo install --path . --quiet
ising --update wolff --l 64 --t-from 2.0 --t-to 2.6 --t-step 0.05 \
  --discard 20000 --measure 100000 --seed 42 --out artifacts/wolff-l64
ising --update wolff --l 32 --t-from 2.0 --t-to 2.6 --t-step 0.05 \
  --discard 20000 --measure 100000 --seed 1042 --out artifacts/wolff-l32
```

② UNDERSTAND

Near T_c single flips need of order L^z sweeps per independent configuration, so running longer loses the race against a larger lattice. The fix is to change the update rule.

The Wolff cluster update [2] flips many spins at once. Pick a site at random. Grow a cluster outward from it: for each neighbour that has the same spin as the site it was reached from, add it to the cluster with probability

$$p = 1 - e^{-2/T}, \quad (16)$$

and never add a neighbour of opposite spin. Repeat for the neighbours of every newly added site until no site is added. Then flip the whole cluster. There is no acceptance test: the flip always happens. Bonds inside the cluster and outside it keep their energy; only the declined bonds at its boundary change, and Equation 16 is exactly the probability that makes acceptance 1 satisfy Equation 6.

The observation interval. One *cluster move* grows and flips one cluster. Record after every move. A cluster's size depends on the configuration; stopping only when accumulated sizes reach L^2 changes the observed distribution. The observation interval must not depend on the sizes just seen. Keeping every move also preserves the correlations needed for error estimates.

Spin-update work. Let c be the number of spins flipped in one move, and $\langle c \rangle$ its measured mean. Express the autocorrelation time in *spin-update work units*, one unit for L^2 flipped spins:

$$\tau_{\text{work}} = \tau_{\text{moves}} \frac{\langle c \rangle}{L^2}. \quad (17)$$

For Metropolis, one unit is L^2 proposed spin flips, including rejections. The conversion compares spin-update counts. It does not measure elapsed-time speedup, since cluster growth and a single-spin proposal have different costs.

The window protocol. The cluster run covers the same 13 critical-window temperatures and both sizes from Part 2. Its observation unit is a cluster move.

Given	Value
Sizes and seeds	$L = 32: 1042; L = 64: 42.$
Temperature grid	2.0 to 2.6 in steps of 0.05, ascending.
Initial state	All-up at the first temperature; reuse the previous final lattice thereafter.
Equilibration	Discard 20000 cluster moves at each temperature.
Measurements	Record every one of the next 100000 cluster moves, including its size.

Output	artifacts/wolff-l64/ and artifacts/wolff-l32/ ; each series.jsonl has one object per cluster move with L , T , sweep , M , E , and cluster_size . Keep the raw rows out of git.
--------	---

Sampler agreement. Both update rules target the same Boltzmann probabilities, so compare $\langle |m| \rangle$ at the same L, T using their sampling errors. For independent runs with standard errors σ_1, σ_2 , use

$$d = \frac{||\langle m \rangle_1 - \langle m \rangle_2|}{\sqrt{\sigma_1^2 + \sigma_2^2}}. \quad (18)$$

Here $|\langle m \rangle$ denotes the measured mean of $|m|$. A difference beyond three combined standard errors calls for more data and an investigation of equilibration and sampling. A small difference supports this observable; it does not establish the full distribution. Part 3's block-length check still applies to each error. A block-length-sensitive error makes the agreement comparison provisional.

- Read Sandvik's notes, section 6, "Cluster algorithms for the Ising model", pages 19–27, and in particular section 6.4, "Wolff algorithm", pages 26–27. Find why flipping the cluster with acceptance 1 satisfies detailed balance, and compare the single-cluster rule with the Swendsen-Wang rule that precedes it.
- Skim Wolff's four-page original [2]. Find the bond probability and the claim about the *dynamic exponent*, the z of L^z .

KEY CONCEPTS

- The *Wolff cluster update* grows aligned clusters with probability Equation 16 and flips them with acceptance 1.
- A fixed interval of one *cluster move* preserves the sampled distribution.
- *Sampler agreement* uses Equation 18 and is provisional when sampling errors remain unstable.
- The critical temperature's *sampling error* excludes finite-size and peak-fitting bias.
- *Spin-update work* converts τ_{moves} with Equation 17 after sampling.

3 VERIFY

Terminal: inside week3/ . Deliverables: the equilibrium-magnetization comparison, the cluster critical-temperature estimate, and week3/evidence/tau-compare.png .

(1) Sampler agreement depends on stable sampling errors.



Draw mean $|M|$ against temperature for size 64 from the metropolis window run and the wolff run in `week3/artifacts/`, with block-bootstrap error bars. Add a second panel of the cluster susceptibility with its five-point fitted peaks and the extrapolated critical temperature. Mark `2.26919`. Save `week3/evidence/magnetization-compare.png`. Discuss uncertainties with me. Commit.

Open week3/evidence/magnetization-compare.png . At $L = 64, T = 2.3$, the Rust reference gives $\langle |m| \rangle = 0.4310$ for the cluster chain and 0.4215 for single flips, a difference of 0.45 combined standard errors using the provisional errors. Compare your runs with Equation 18 and inspect both error bars across block lengths. Report agreement when both errors are stable in Part 3's sense, within a tenth of their mean across the three block lengths, and the comparison is at most 3. If either error remains sensitive to block length, report "agreement provisional", as for the reference runs here. Report a value above 3 as a discrepancy needing investigation.

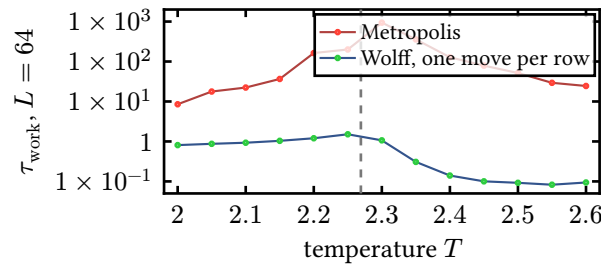
- (2) **The cluster critical temperature has a small sampling error and a remaining bias.** Read the fitted peaks and the extrapolated temperature in the right panel of week3/evidence/magnetization-compare.png . The Rust reference peaks are 2.3492 and 2.3117, giving $T_c = 2.2741$, within 0.22% of 2.26919. Your central estimate must satisfy the same 2% comparison as Part 2. The Rust reference's block bootstrap gives a sampling error of 0.0007 at each of the block lengths 2000, 4000, and 8000 cluster moves, thirteen times smaller than the single-flip bar and small against the 0.0049 distance from the exact value. Explain why more independent samples reduce sampling error while the two-size extrapolation and five-point fit can retain a systematic bias.

(3) Cluster updates reduce the work per independent sample.



Write `week3/scripts/compare.py`: draw work-normalized autocorrelation time against temperature for size 64 from the metropolis window run and the wolff run in `week3/artifacts/`. One cluster move costs `mean cluster size / L^2` sweeps of spin updates. Save `week3/evidence/tau-compare.png`. Run it. Discuss uncertainties with me. Commit.

Open `week3/evidence/tau-compare.png`. Use Equation 17 for the cluster curve and sweeps for Metropolis. Near $T = 2.3$, the cluster value should be smaller by a factor in the hundreds. The Rust reference records $\tau_{\text{moves}} = 4.736$ and $\langle c \rangle = 921.6$, giving $\tau_{\text{work}} = 1.066$. Against the single-flip value 701.8 of the same window run, this is about 660 times less spin-update work per independent $|m|$ sample. Report your measured ratio; do not require it to equal this reference value.



Work-normalized autocorrelation times. Both series use the Rust reference implementations. The cluster series records each move and converts time afterward.

What the sampler omits. The sampler you built covers one model with one order parameter and two update rules. It omits histogram reweighting, Binder cumulants, parallel tempering, and the checkerboard decompositions that put single-flip updates on a GPU. Serious work uses packages such as ALPS [3], with Newman and Barkema [4] and Landau and Binder [5] as the textbooks. Build it once anyway: the sampler, the sampling- error estimator and the two-size extrapolation are what those packages hide; this week you build all three, with the systematic bias stated but not measured.

Wrap up the work

Goal: a README anyone can follow to regenerate the week's evidence.

The committed tree contains the code, recording, and visual evidence:

```
week3/
  Cargo.toml, Cargo.lock, ising.design.toml, README.md
  src/, tests/
  scripts/                                ← peaks.py, errors.py, compare.py, and the plot
scripts
  spins.jsonl
  evidence/
    boltzmann.png, magnetization.png, susceptibility.png
    trace.png, acf-binning.png, tau.png
    chi-bootstrap.png, magnetization-compare.png, tau-compare.png
    viewer-T1.8.png, viewer-T2.3.png, viewer-T3.0.png
```

Keep `week3/evidence/` and the viewer recording tracked; `artifacts/` stays in the local clone. Keep each committed file below 5 MB.

Your `week3/ising.design.toml` is the contract Part 1 gave you. The reference copy ships with the week: [download week3-ising.design.toml](#). Download it and compare it with yours; the two files must match section for section.

At the exercise repository root, send:



Write `week3/README.md` as a guide to regenerate every committed file from a clean clone: the install command, every `ising` run with its values, then each script with the file it writes, in order. List every file in `week3/evidence/` beside the command that produces it. Discuss uncertainties with me. Commit.

Follow the README yourself, from the top, in the second terminal; every command must run as written and every listed file must appear. Then commit and push:



```
git add -A &&
git commit -m "week 3 evidence" &&
git push
```

Show the instructor the committed evidence and explain the limits of your critical-temperature estimate at the next session. Nothing is submitted.

Challenge: beat Wolff at the critical point

Neural samplers. The update you built in Part 4 is the state of the art of 1989. Since 2019 a line of work has replaced the chain's proposal with a neural network that draws a whole lattice at once, instead of one spin or one cluster. Four families are in use. An *autoregressive network* writes the spins one after another, each conditioned on the ones already written. A *normalizing flow* maps easy random numbers through an invertible network into a lattice. A *transformer* is a network whose layers let every site attend to every other. A *diffusion model* starts from noise and removes it in steps. None of these is exact on its own, so the samples are corrected in one of two ways. The first keeps a Markov chain and applies the acceptance test of Equation 7 to the proposed lattice, which gives a *Metropolis-corrected neural chain*. The second keeps every sample and weights it by the ratio of the Boltzmann probability to the probability the network assigned it, which is *neural importance sampling*; its figure of merit is the *effective sample size*, the fraction of drawn samples that count as independent. The promise of both is a proposal whose cost does not grow at T_c , and the free energy as a by-product, which no Monte Carlo chain gives you.

The published record on the two-dimensional Ising model at T_c , checked on 2026-09-06, is short. The largest Metropolis-corrected neural chain is $L = 128$, and it reports only a relative autocorrelation statement [6]. The largest importance-sampled run is $L = 180$, with an effective sample size of 0.59 after 192 hours of training, no Metropolis step, and the authors' own verdict that neural samplers are not competitive with cluster algorithms [7]. No paper has fitted a dynamic exponent for a neural chain; an absolute τ_{int} exists only at $L = 16$ [8]. No paper has counted training plus sampling cost against Wolff in two dimensions at any size, and the one such count, in three dimensions at $L = 8$, came out about $70 \times$ in Wolff's favour [9]. Critical slowing down does not disappear in these methods. It moves into training, where the learning time of a one-layer score network grows as L^2 at criticality [10], and the importance weights can look healthy while whole modes of the distribution are missing [11].

The benchmark. Temperature is written here as the *inverse temperature* $\beta = 1/T$, the convention of this literature, so Onsager's point is $\beta_c = \ln(1 + \sqrt{2})/2$.

Row	Requirement	Never reached because
Size	$L = 256$ at β_c , on the grid and conventions of Part 2.	The largest corrected chain is 128; the largest reweighted run is 180.
Exactness	A Metropolis-corrected chain with acceptance ≥ 0.5 at $L = 256$; free energy within 10^{-5} relative of the exact finite-lattice value [12]; the energy histogram passes a χ^2 test against the exact finite-lattice energy distribution [13].	The histogram test has never been applied to a neural sampler at this size, and reweighting-only results carry no acceptance.
Dynamics	τ_{int} of $ m $, E and χ at $L = 32, 64, 128, 256$, with error bars from the independence-sampler estimators [8], and a fitted exponent z' with an error.	No z' exists.

Cost	Total cost per independent sample at $L = 256$, training amortized over the run, at or below Wolff's on the same hardware, reported in wall-clock and in floating-point operations.	Never counted in two dimensions; $70 \times$ the wrong way in three dimensions at $L = 8$.
Reproducibility	Seeds, the usage lines, released code, and a check that fails: a deliberately mode-collapsed model must fail the histogram test.	No published neural-sampler result ships a check with teeth.

Where to start. The tool of this week is the reference implementation. Wolff gives the ground-truth chain at any L in minutes, so every neural number has something to be measured against. `scripts/errors.py` already holds the τ_{int} estimators and grows the independence-sampler ones. Two more printed comparisons join: the exact finite-lattice free energy and the histogram test. The artifact contract stays as it is, so the instructor's gate keeps working on the new sampler. Public code to build on: the hierarchical autoregressive networks [6], the multilevel sampler [14], the neural cluster updates [15], and the exact planar-Ising sampler [16] as a second ground truth. Read [17] for the estimators and [18] for why training is the cost.

What counts. Clearing every row of the table is the bar nobody has reached. Clearing Dynamics and Reproducibility with the cost curve, even when Wolff wins, is a new result, and a scaling study that stops below $L = 256$ with a stated reason is a valid negative outcome.

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